

Urban energy modeling and calibration of a coastal Mediterranean city: The case of Beirut

Alaa Krayem^{a,b,c}, Ahmad Al Bitar^b, Ali Ahmad^d, Ghaleb Faour^a,
Jean-Philippe Gastellu-Etchegorry^b, Issam Lakkis^d, Jocelyne Gerard^e, Haitham Zaraket^c,
Aram Yeretian^d, Sara Najem^{a,*}

^a National Council for Scientific Research, Beirut, Lebanon

^b Centre d'études spatiales de la Biosphère, Toulouse, France

^c Lebanese University, Beirut, Lebanon

^d American University of Beirut, Beirut, Lebanon

^e Saint-Joseph University, Lebanon

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ABSTRACT

Urban expansion, driven by population and economic growth, has been a major contributor to increased levels of energy consumption across the globe. Beirut, Lebanon's capital, is no exception in facing a surge in its demand for electricity as it expands. However, with frequent power outages, underpinning a largely problematic power sector, Beirut's demand for electricity is becoming a real hurdle that impedes the city's economic growth and development. This paper introduces a near-city-scale building energy model, BEirut Energy Model BEEM, which estimates the building stock's electricity consumption in two different districts in Beirut. The methodology uses rule-based expert data for an archetypal classification of the buildings based on their functions and periods of construction with their corresponding attributes including the number of floors, number of apartments, and bimonthly electricity consumption to generate a 3D model for 3630 buildings coupled to the hourly weather conditions and topographic map, which is then simulated in EnergyPlus. The predicted consumption of 2311 buildings is then calibrated with actual available metered data, to adapt the model to Beirut's occupancy and users' behaviors. Calibrated results are mapped to reveal the spatiotemporal distribution of energy peak demands which provide insights for future interventions. An analysis of the spatial distribution of electricity use demonstrates a spatial clustering that underlies urban energy demand which can be used for smart grid zoning.

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1. Introduction

In 2016, 54.5% of world's population lived in urban areas. By 2030, cities are projected to house 60% of the global population [1]. Among the climate challenges in urban environments is the urban warming measured in numerous cities across the world in addition to the Green House Gas (GHG) emissions both linked to the increased urbanization level [2,3]. For the aforementioned challenges, a relevant understanding of cities' metabolisms is pivotal in developing energy efficient strategies [4]. Of particular interest in this study is Beirut administrative district of Greater Beirut, Lebanon's capital, which held 50.7% of the urban population of the

country and 44.6% of the total population in 2016 [1], and consumes 12% of the total national energy produced while it only covers 0.2% of the country's total area. Its tertiary sector (commercial sector, public offices, hotels and hospitals) and residential sector account for 73% and 26% of its electricity consumption, respectively. These represent 39% and 14% of the electrification GHG emissions, respectively [5]. On the other hand, the generated electricity does not cover the country needs, leading to 12 h of outages in some parts of the country, which is compensated with back-up highly polluting private generators. This situation challenges Lebanon's pledge to reduce its GHG by 30% by 2030 with 2015 being the base year. The country is then facing a significant challenge to manage its energy sector and integrate renewable energy. In order to ensure the optimization and the prioritizing of the energy resources and the energy conservation measures (ECM) to be applied, programmatic decision making or energy management

* Corresponding author.

E-mail addresses: snajem@cnrs.edu.lb, sara.najem@mail.mcgill.ca (S. Najem).

is required. It consists of planning, implementing and monitoring energy supply in efficient manners to reduce its waste and consumption [6]. Therefore, developing an urban scale energy model is essential to Beirut's resources and energy supply management. Such models enable the understanding of how a city's energy system changes over space and time under these interventions and are crucial to support the decision making process. Solutions range from the individual building energy models (BEM) [7] to the urban scale or the so-called Urban Building Energy Models (UBEM), which allow the assessment of building to building interactions (shading, heat exchange, etc.), and of building to other urban components interactions such as the urban heat island and traffic [8]. City scale energy balance models [9–11] based on top-down approaches are used for climatic modeling but they do not provide the needed details to test innovative scenarios at building scale. On the other hand, bottom-up physical simulation models were introduced as effective simulation tools to model the impact of the urban context on buildings energy demand [12–16]. BEM and UBEM need to be reliable and adaptable in that they should provide accurate estimates of the buildings' energy performance. However, discrepancies often occur between the predicted model and the actual metered building energy use, mainly due to inputs' inaccuracy/uncertainty in model parametrization and structure [17]. Hence, the calibration of energy models is critical to achieve a confidence level in model predictions and encourage their adoption. Such calibrations range from deterministic manual iterative tuning [17,18] to automatic calibration process using optimization techniques, machine learning algorithms or Bayesian calibration [19]. Once calibrated, UBEM can be useful to estimate impacts of new technologies and policies where no measured data is available and to reveal unknown occupant patterns.

This paper introduces a new methodology for city scale UBEM implementation applied to the case of BEirut Energy Model (BEEM), a near-city-scale 3D energy model. It accounts for the buildings architectural characterization and human behaviors to estimate the buildings energy performance. A priori, simulations were performed by using ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers) occupancy schedules [20]. As an application of the energy models with standardized parameters to Mediterranean cities like Beirut, the main goal of this paper is to provide insights regarding the adaptation of these energy models to other cities having different occupants lifestyles and behaviors. The urban morphology, building types and configuration as well as construction materials in Beirut are similar to that of other Mediterranean cities, which makes the presented work suitable for replication in other cities within the region [21,22]. By adapting our model's configurations to better represent the building-level bimonthly energy consumption provided to us by the Electricité du Liban (EDL), the sole provider of electricity in Beirut, we recover the spatiotemporal distribution of occupancy schedules, which is a missing and vital component in the optimization of energy resources. The multi-scale nature of the model allows the determination of sustainable solutions ranging from building level to the city level. It also serves as a valuable tool for energy resources management and for improving existing energy policies. Another characteristic of the energy model for Beirut is its capability to project the energy consumption under normal conditions, that is with no rationing and outages. Currently, estimates of demands do not account for the suppressed amount of electricity, since during outages, occupants modify their behaviors and alter their energy consumption patterns and preferences.

This study begins with a review of existing urban energy models field and a description of the workflow adopted to build the energy model in Section 2. Simulation results are presented in

Section 3 followed by a discussion of the findings and the model capabilities in Section 4.

2. Methods

2.1. Urban scale energy models

Two main approaches have been used to model energy use patterns in urban context: *top-down* and *bottom-up* models [23]. Top-down models are principally applied to explore the inter-relations between the energy sector and the socio-economic factors [23] or to model the climatic conditions [24,25]. However, top-down models are unsuitable for interventions and technological studies [26]. In contrast, bottom-up models provide estimates of individual end-uses. Results are aggregated according to their impacts on energy use and the challenge remains in extrapolating them to regional or national level. This detailed approach allows for improvements and technological studies. Bottom-up models, are broadly divided into *statistical* and *engineering* models [27]. Statistical models use existing data and incorporate occupant behavior by attributing energy consumption to end-uses [23,28–30]. However, these models that are based on training over annual metered data are unable to predict energy use in monthly or hourly time steps or to simulate the combined impact of several energy efficiency measures in buildings [12]. On the other hand, engineering models are highly accurate if provided with correct inputs. They apply thermodynamics and heat transfer equations in the building energy simulation [31–35]. However, a detailed physical description of building properties and systems and precise weather characteristics are essential for such approach [36]. A bottom-up approach was used to assess Lebanon's energy budget from 2010 to 2015 [37]. For the building sector, the calculations were based on the constructed area per building type, the climatic zone, the occupancy rates and the energy demands per end-uses per building type defined by a previous study [38]. The latter study results were obtained after simplified calculations conducted for a Business as Usual Case (BAU) under coastal climate and with assumptions on the non-geometrical properties based on expertise. However, both studies, [37,38], did not account for the different properties of buildings envelope neither their shading effect and consequently no spatiotemporal analysis was carried out.

Generally, UBEM applies heat transfer equations in and around the buildings, where each building is represented as an individual 3D dynamic thermal model. Furthermore, it can be combined with energy simulation tools to provide hourly energy predictions [12]. Given the massive amount of data for potentially thousands of buildings, the energy modeling workflow of an UBEM requires extensive effort and time resources. Assembling, managing, and automating the workflow is therefore essential. For this purpose, the building stock is divided into archetypes to reduce complexity and computation requirements [12,39]. A typical UBEM can be constructed using a set of tools comprising a Geographic Information System (GIS) for buildings' geo-morphological properties information, a CAD environment such as Rhinoceros 3D [40], and an energy simulation engine such as EnergyPlus [34]. Broadly, the workflow consists of generating the archetypes based on key parameters such as the year of construction and buildings' types. Then the building's footprints are extruded to create the three-dimensional form, where the heights are defined by product of the number of floors and the characteristic floor height of the year of construction, then dividing the floors into sub-zones, generating windows and balconies with the corresponding window to wall ratio and assigning the specific thermal properties all based on the building's archetype. Shading surfaces are determined and each building is then represented by a thermal mode.

Table 1
Original variables datasets and the selected variables for BEEM.

Data Origin	Form	Type	Original field name	Designation
USJ survey	Shapefile	Polygon	Building footprint	–
			Enquete ID	Building ID
			Annee	Year of construction
			Fonction	Function
			NbreEtag	Number of floors
			NbreApparts	Number of apartments
			area	Area of a floor
EDL	Shapefile	Numeric values	perimeter	Perimeter of a floor
			Months 2	Electricity consumption in January and February
			Months 4	Electricity consumption in March and April
			Months 6	Electricity consumption in May and July
			Months 8	Electricity consumption in June and August
			Months 10	Electricity consumption in September and October
			Months 12	Electricity consumption in November and December

2.2. 3D energy model workflow

2.2.1. Data collection

To create the model dataset, three types of data essential for the model characterization were collected from a range of sources. They are: weather data, buildings geometrical properties and buildings non-geometrical properties. Hourly weather data is obtained from the Beirut International Airport weather stations, which is located along the Mediterranean coast in the southern suburbs of the city of Beirut. A topographic map (scale 1:20000) with 10 m contour lines, provided by the National Council for Scientific Research CNRS-Lebanon, was used to create the digital elevation profile of Beirut. Beirut has around 17,000 buildings. Detailed information about their characteristics is, however, largely incomplete. Information about 7122 buildings in different areas within Beirut were surveyed by the University of Saint Joseph USJ. The dataset, associated to the GIS in the form of attribute tables, labels each building by its footprint and a certain number of entry fields summarized in Table 1.

2.2.2. Data processing and archetypes generation

Bi-monthly electricity consumption of a set of 2311 buildings for the two districts was provided by EDL and added to the updated dataset. Buildings near the shore (up to 1 km inland) have in general their facades face west and/or north with large windows overlooking the sea. To take into consideration this feature, we labeled these buildings. Moreover, a comprehensive survey was used to label glazed buildings as “glazed”.

The buildings' plots were cross-checked with a Pleaide satellite image. Buildings with incomplete information were excluded. This reduced the number of buildings from 7122 to 3630 distributed over Beirut as shown in Figs. A.1 and A.2 in the Appendix A.

To overcome the complexity of dealing with a large dataset of distinct buildings, archetypes were generated. An archetype is a set of geometrical and non-geometrical properties that characterize a building's thermal performance. Non-geometrical properties include the thermal resistance of the construction materials, occupancy schedules and appliances densities [13]. Archetypes generation consists of two steps [12]: the *segmentation* of buildings with similar properties based on key parameters such as the type, age, shape [41–44], and the *characterization* of the thermal properties for each archetype. This information was gathered from buildings audits [45], existing database [46–48] or local expertise and previous literature [41]. In this study, the buildings were grouped based on two parameters: the building's type and the year of construction based on a historical architectural analysis of the buildings. The building type helps in setting a building's occupancy patterns and determining its internal heat loads, while its year of construction informs about construction material and sequence of layers

of each building component which determine its thermal and optical properties. For the specific case of Beirut, five distinguished construction periods were identified in the literature: 1900–1923, 1924–1940, 1941–1960, 1961–1990, and 1991 and on [49]. Regarding the type, the buildings were grouped in six classes: residential, mixed, hospitals, schools and governmental buildings. In total, $6 \times 5 = 30$ archetypes were generated. Thermal properties were acquired from the Technical Guide for the application of the Thermal Standard for Buildings in Lebanon [50], published in 2005. Missing information was obtained from ArchSim default library [51] and online libraries [52]. A priori, non-geometrical properties, including light and equipment loads intensities and usage schedules were set by referring to ASHRAE standards [20]. HVAC systems performance coefficients were obtained from Ref. [38]. Finally, ArchSim library was updated with the missing properties as inputs for the simulations. Fig. 1 summarizes the adopted workflow.

2.2.3. Thermal model generation and execution

After preparing the data, buildings were divided into floors and each was assigned an archetype. The geometry modeling process is shown in Fig. 2. The buildings footprints and the topographic map contours were imported to Grasshopper, a plugin of Rhinoceros 3D [40]. After generating the Digital Terrain Model (DTM) using Delaunay triangulation, buildings footprints are projected on the DTM surface. The polygons were then extruded to the buildings heights which, estimated as the product of the number of floors and the floor height for the corresponding year of construction. Using the ArchSim tool, each building was divided into its number of floors, each represented by a single thermal zone. No further zoning was applied in this study. Next, windows were generated on free facades, which resulted from excluding adjacent walls and those whose neighboring wall is less than 3 m apart. To have a more realistic representation of the buildings, the number of windows per facade was restricted by its width and the Window to Wall Ratio WWR given by orientation and year of construction. In this respect, windows were generated with a spacing of 5 m for residential buildings and 4 m for non-residential buildings. The WWR was primarily identified according to the year of construction. Values ranged between 0.15 in 1900–1923 and up to 0.7 after the 1990s. However, if the building overlooks the sea, the WWR was increased to 0.7 to take into account the north and west highly glazed facades. In addition, fully glazed buildings and first floors of mixed buildings, considered as retail shops, were assigned a 0.95 and 0.8 WWR, respectively. Another feature of BEEM is the possibility to include balconies, which are added to free walls of residential floors having an area greater than 15 m².

Potential shading surfaces from neighboring buildings were determined in the 3D model based on a two-step process. First, within a radius of 300 m, each building (of height H) that is $3.78 H$

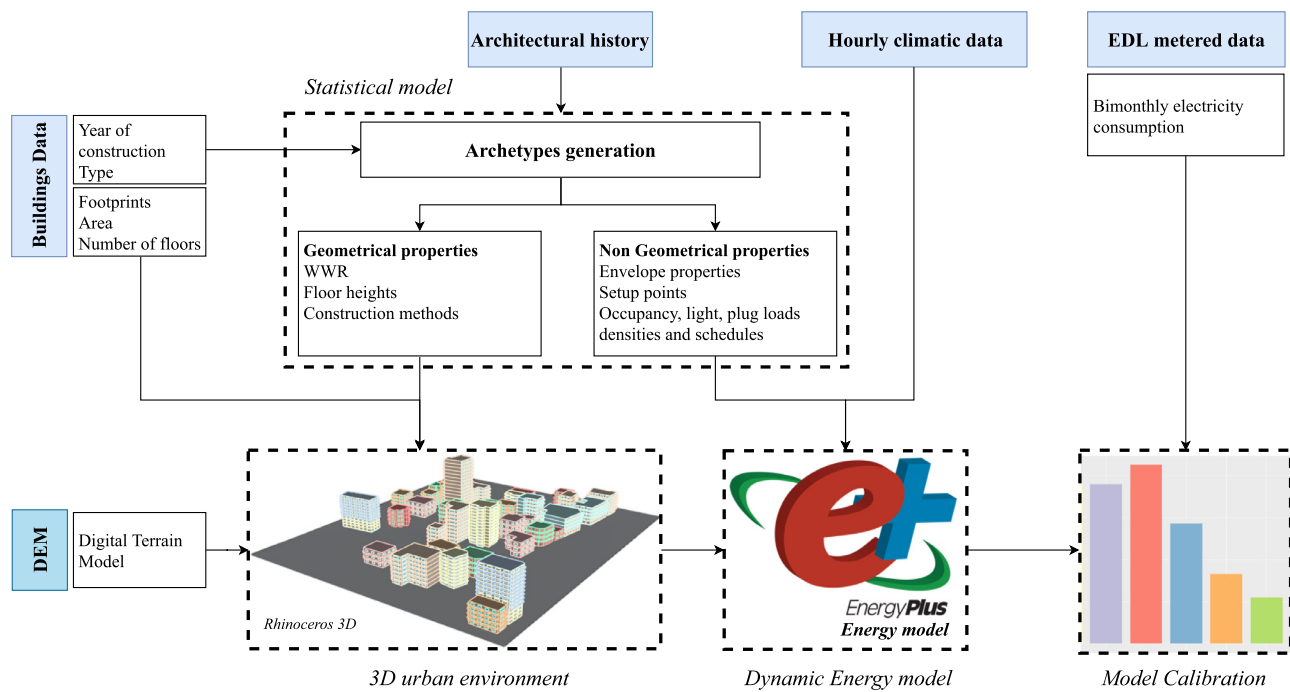


Fig. 1. Beirut Energy Model BEEM methodology flowchart.

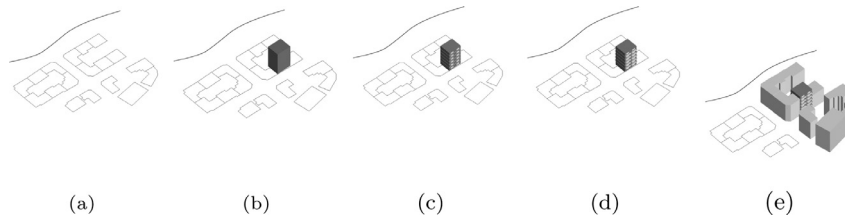


Fig. 2. Fig. (a) shows the DTM creation, followed by a simple extrusion shown in Fig. (b), then windows and balconies generation in (c) and (d), respectively, and finally the adjacency and shading are shown in (e).

away from the target building is considered as a building that casts a shadow. The multiplicative coefficient 3.78 was calculated as in [53], based on a sun angle of 14.8° , corresponding to 80% of the time between 5:30 am to 8 pm in Beirut. To further reduce the number of shading surfaces, a ray tracing algorithm was used to detect only the neighborhood buildings' surfaces that obscure the solar radiation. Once the 3D model was completed, each thermal zone was assigned the simulation parameters from the buildings database.

A thermal model was generated for each building using EnergyPlus, using ArchSim tool. The runs were distributed on four virtual machines created on Microsoft cloud service Azure. For the calibration process, the results were then processed and compared to the EDL data, when available, as discussed next.

2.3. Model calibration

As previously mentioned, bimonthly EDL data was available for a certain number of buildings. Outliers were removed using density-based spatial clustering of applications with noise (DBSCAN) and autoencoders algorithms, based on the buildings' area, number of floors, type, year of construction and EDL consumption. Once outliers were identified, the remaining buildings were clustered based on their simulated electricity consumption and actual electricity consumption (EDL). This clustering helps identify buildings with similar occupancy schedules.

The offset between EDL values and those predicted by the model can be mainly attributed to behavioral patterns, HVAC sys-

tems and systematic errors. Hence, any correction to be applied must be consistent with the above categorical errors, adaptable to the city's context and justified at the same time. For this purpose, the following interventions were implemented:

- **Systematic errors:** 3 hours long electricity cutoff in Beirut accounts for a reduction factor of $1/8$, when averaged across the year. Hence, the simulated electricity consumption was reduced by that amount. In addition, errors related to numerical algorithms were eliminated. These errors are the results of the Energy Plus scheme to adjust the zone temperature to meet the desired value defined by the control system set points.
- **HVAC systems architecture:** cooling in Beirut relies mostly on unitary cooling units associated to given areas. The units are rarely running at the same time. To this end, we assume that only 50% of the floor area is cooled or heated at a given time, so that the simulated cooling and heating consumption are reduced to the half. In addition, when the outdoor temperature was less than 20°C , the cooling loads were nullified.
- **Energy use and occupancy profiles:** after applying the previous corrections, we assumed that the remaining discrepancies were caused by occupancy profiles, linearly altering the total bimonthly electricity consumption by a certain factor. This assumption is justified in Section 3.2. Therefore, the calibrated energy consumption for each building was obtained as the multiplication of model estimated energy consumption and the averaged multipliers of its cluster.

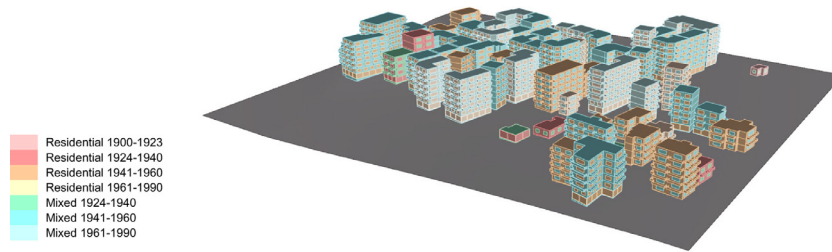


Fig. 3. Sample of the generated 3D model of buildings in the Bachoura area, Beirut, Lebanon.

3. Results

3.1. Archetypes distribution and 3D model

The USJ data covered two districts in Beirut as shown in Fig. A.1 of the Appendix A. Attributing archetypes to buildings shows that residential and mixed buildings are the two main types in the two districts with 57.9% and 38.4%, respectively in district A and 62.7% and 35%, respectively in district B, as shown in Fig. A.1 of the Appendix A. Almost 40% of the buildings in both districts were built between 1941 and 1960 (the golden period) while 34% and 23% were built between 1961 and 1990 in district A and district B, respectively, as shown in Fig. A.2. The main difference between the two districts is the buildings facing the sea in the north and the west of district A. All of these buildings are residential and mixed. In addition, most schools, hospitals and governmental buildings are within district B.

The 3D model of a neighborhood in the Bachoura area in district B is shown in Fig. 3. The figure illustrates the use of archetypes to generate realistic 3D models from the thermal point of view, including windows and balconies.

3.2. Data processing and model calibration results

After generating the thermal model, annual hourly loads from equipment, lighting, domestic hot water, cooling and heating were aggregated into simulated electricity consumption for residential and mixed buildings. In hospitals, the electricity consumption does not account for the heating and hot water demands since they are met by steam boilers. As for schools and governmental buildings, hot water is not available in general and therefore not accounted for. The electricity consumption profiles for residential and mixed buildings are described in Section 3.3. Note that other building types were excluded from the calibration process as we will see in the next paragraph.

Following the interventions implemented in Section 2.3, 31% of the buildings had to be removed due to missing EDL data entries. Note that all governmental buildings and hospitals lacked EDL data and were, as such, eliminated at this stage. To ensure consistency of the calibration process, schools, representing 1.3% of the remaining dataset, were removed because their schedules are different than those of residential and mixed buildings.

After applying the noise detectors algorithms (dbscan and autoencoders), 64 buildings in A and 117 buildings in B were recognized as noises and therefore eliminated. The remaining data included 481 buildings in A and 1830 in B, all residential or mixed. The average EDL electricity consumption per floor area for the final set of buildings was 69 kWh/m² and 83 kWh/m² for residential and mixed buildings, respectively in district A, and 62 kWh/m² and 68 kWh/m² for the same buildings types, respectively in district B. The simulated consumption was greater than the actual consumption by a minimum of 2.5% and a maximum of 236% in district A, and a minimum of 2.85% and a maximum of 262% in district B.

Model calibration is done by applying a k-means clustering to the ratio of EDL to predicted consumptions. This led to the iden-

tification of four clusters of buildings in each district, as shown in Figs. 5a and 6a. The number of clusters was defined based on an optimization procedure for each district. The model overestimated the electricity consumption in most buildings, mainly grouped in clusters 0 and 1 in both districts. Clusters 3 grouped buildings with EDL to simulated electricity consumption ratio ranging between 0.6 and 1.3 in district A and between 0.5 and 1.5 in district B. The consumption of only 21 and 34 buildings were underestimated in districts A and B, respectively.

No specific relation between the clusters distribution and the buildings coordinates could be concluded. Moreover, the clusters distribution was shown to be independent of the buildings' type, year of construction and archetypes.

Thus, energy use and occupancy profiles can be considered as the main source of the disparities between actual and predicted consumption. More specifically, maximum occupancy corresponds to buildings whose electricity consumption is the highest compared to their counterparts with the same number of floors. Fig. 4 shows a strong correlation between the maximum actual consumption and the predicted consumption of each building normalized to its actual consumption, which validates our hypothesis. As a result, we assumed that the discrepancies in electricity consumption between EDL and our model were related to usage profiles and hence, we adjusted the consumption of each building according to its cluster coefficients representative of specific energy patterns. Table 2 summarizes the clusters multipliers for both districts.

The calibrated results, plotted against EDL data in Figs. 5b and 6b, show that the buildings consumption of each cluster are more grouped around the identity axis.

3.3. Loads profiles

Comparison of the electricity consumption from EDL data with model predicted and calibrated output at a bimonthly resolution for the buildings in both districts is shown in Fig. A.3 in the

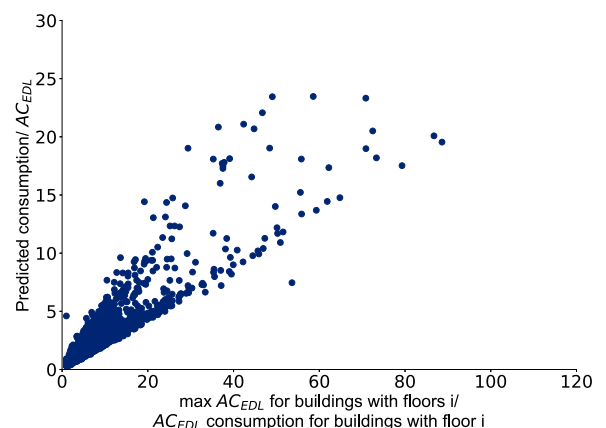


Fig. 4. Correlation between the ratio of the maximum EDL consumption by floor to EDL consumption, and the ratio of the predicted consumption to EDL consumption.

Table 2

Multipliers used to equate the bimonthly profile consumption with the cluster average consumption.

		Jan-Feb	Mar-Apr	May-Jun	Jul-Aug	Sep-Oct	Nov-Dec
District A	Cluster 0	0.35	0.351	0.313	0.271	0.274	0.391
	Cluster 1	0.793	0.82	0.743	0.618	0.665	0.933
	Cluster 2	1.556	1.618	1.38	1.136	1.165	1.779
	Cluster 3	3.18	3.01	2.54	2.07	2.219	3.16
District B	Cluster 0	0.326	0.306	0.236	0.199	0.216	0.408
	Cluster 1	0.697	0.764	0.689	0.610	0.629	0.945
	Cluster 2	1.38	1.483	1.310	1.139	1.187	1.815
	Cluster 3	2.307	2.767	2.592	2.186	2.346	3.331

Table 3

Comparison between the energy model outputs and the actual metered data before and after calibration.

District A	Jan-Feb	Mar-Apr	May-Jun	Jul-Aug	Sep-Oct	Nov-Dec	Yearly
EDL (Mwh)	6,302	5,675	6,674	7,506	6,650	6,413	39,220
Predicted (Mwh)	13,444	12,282	18,144	24,788	20,522	13,289	102,469
RMSE (Kwh/m ²)	15	12.5	18.8	27.5	22	13.1	105.4
Calibrated (Mwh)	6,250	5,733	6,737	7,473	6,693	6,545	39,433
RMSE (Kwh/m ²)	6.5	4	4.6	5.2	4.8	4	23.7
District B	Jan-Feb	Mar-Apr	May-Jun	Jul-Aug	Sep-Oct	Nov-Dec	Yearly
EDL (Mwh)	15,631	14,017	16,993	20,229	17,702	18,013	102,585
Predicted (Mwh)	40,989	36,774	53,977	73,844	60,904	39,632	306,119
RMSE (Kwh/m ²)	17.8	12.5	19.6	29.1	23	13.4	112
Calibrated (Mwh)	15,156	14,237	17,017	19,913	17,602	17,898	101,823
RMSE (Kwh/m ²)	4.7	3.4	4.4	5.1	4.4	4.5	22

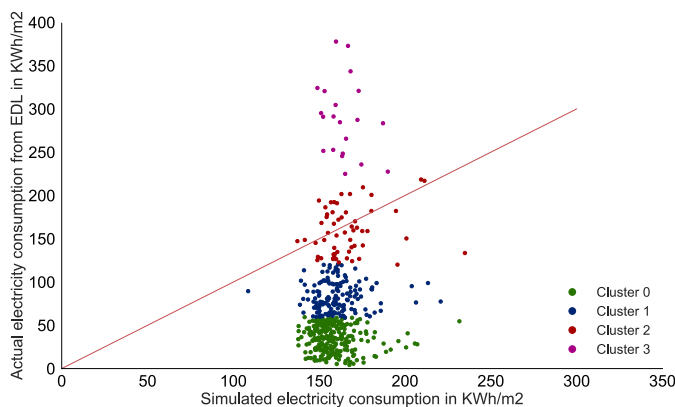
Appendix A. Table 3 illustrates the differences between the model outputs and EDL data before and after calibration.

We found out that the monthly variation of the energy consumption predicted by the model before and after calibration follows a similar trend to that of actual data: March and April were the months of least consumption, while peaks were detected during the hot months (July and August). The Fig. A.3 also depicted the overestimation of the electricity needs variation over the year. The increase in energy demand between the less consumer months (March and April) and the most consumer months (July and August) was predicted by 97% and 107% for residential and mixed buildings, respectively in district A, while in reality, it was only of 32% for both buildings types in district A. Very simi-

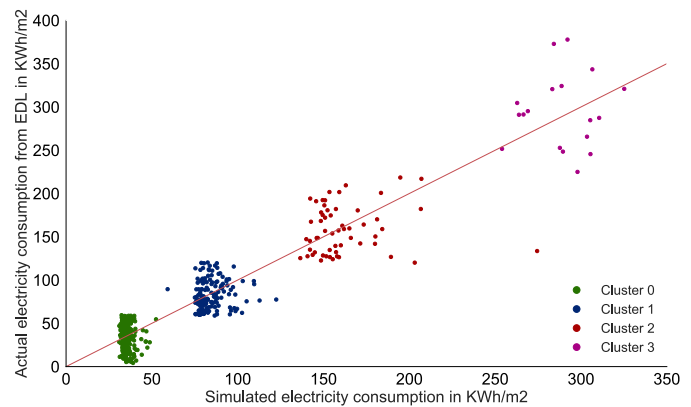
lar trends were found in district B, except that the actual increase was of 42% between the second couple and the fourth couple of months.

Figs. A.4 and A.5 in the Appendix A show the end-use apportionment of the bimonthly electricity consumption after calibration for residential and mixed buildings in both districts. Table A.1 summarizes the percentage of energy consumption for each end-use in the two buildings types.

By combining the model output with GIS mapping techniques, a spatiotemporal distribution of the energy consumption is obtained, as shown in Fig. A.6. Note that buildings with the highest consumption were built between 1941 and 1990. This we suspect to be related to their high occupancy.

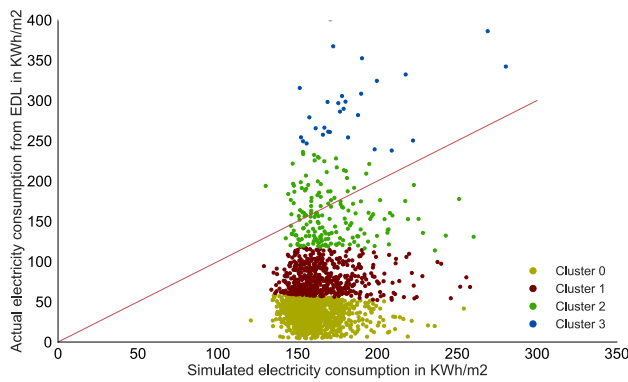


(a) Default EPlus configuration: full occupancy with ASHRAE standards

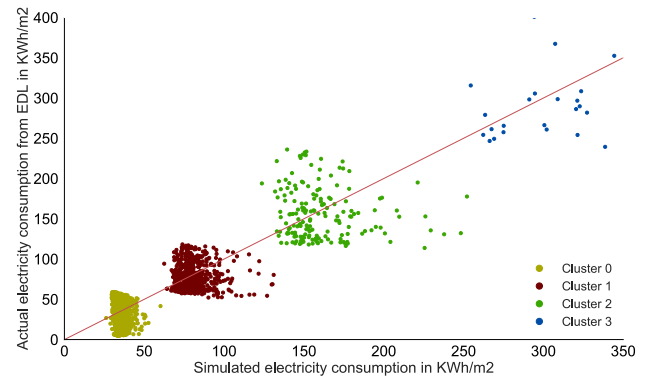


(b) Adapted EPlus results to Beirut occupancy and user behavior linked to socioeconomic variability.

Fig. 5. Scatter plots of buildings consumption in District A for (simulated electricity consumption in kwh/m²) in x axis versus (actual electricity consumption in kwh/m²) in y axis after clustering.



(a) Default EPlus configuration: full occupancy with ASHRAE standards.



(b) Adapted EPlus results to Beirut occupancy and user behavior linked to socio-economic variability.

Fig. 6. Scatter plots of buildings consumption in District B for (simulated electricity consumption in kWh/m^2) in x axis versus (actual electricity consumption in kWh/m^2) in y axis after clustering.

3.4. Spatial autocorrelation

In order to examine the spatial variation of the electricity consumption of residential and mixed buildings, we used global Moran's I to evaluate the spatial autocorrelation. Moran's I assesses if the electricity consumption exhibits certain spatial patterns or it is random. To determine to which extent a variable is dependent on the values observed at neighboring locations, we implement an incremental spatial autocorrelation. The analyses were based on the energy consumption per building. By comparing EDL autocorrelation results with the calibrated model results, they both appeared to be the strongest at 60m in district A. On the other hand, predicted energy consumption in district B showed similar results as the EDL data with the strongest correlation being at 40m for all end-uses. The p-value was less than 1% which indicates that the distribution of the energy consumption is not random.

3.5. Archetypes spatial correlation

In order to determine if buildings of different archetypes are linearly related in terms of electricity consumption, the Pearson coefficient was used. The value of the Pearson coefficient ranges from -1 to 1 . When it approaches 0 , the correlation of the two variables is weak. If the correlation factor is near 1 or -1 , a strong correlation exists between the two variables. The Pearson coefficient is used to analyze the relation between a building's energy consumption with the neighboring buildings. However, the definition of neighbors buildings can vary. Therefore, we applied the correlation analysis to a radius of 1500m with a step of 50m. When no buildings of the corresponding archetype reside within the given distance or no sufficient information can be deduced, null values were forced. In district A, correlation was only found between the electricity consumption of mixed buildings from the third period of construction and residential buildings from the second period on one hand and mixed buildings from the fourth period on the other hand, as shown in Fig. A.7. The correlation was quite moderate in both cases and was lost at 200m in the first case and at 100m in the second. In district B, a significant correlation was found between residential buildings of the fourth period and mixed buildings from the third period of construction ranging from 0.75 at distances less than 50m to 0.25 at 100m. Furthermore, mixed buildings from the second period of construction had

positive correlation with mixed and residential buildings from the third and fourth periods of construction.

3.6. Temporal correlation

In order to determine the temporal variation of the electricity consumption, we applied linear correlation analysis between the energy consumption of the different months by the use of the Pearson coefficient. As in the previous section, null values were forced when there was no consumption of electricity. A significant correlation characterized the electricity consumption of cooling equipment in hot months. For March and April, the correlation factor was around 0.85 and 0.95, respectively in district A and 0.7 and 0.8, respectively in district B with respect to months between May and October. The cold months were also highly correlated in terms of heating.

4. Discussion and conclusion

The discrepancies between BEEM predictions and actual consumption data can be associated to the number of appliances, and occupancy schedules related to occupant's comfort and income level. In this context, most residents in Beirut tend to reduce their electricity usage for heating in winter. Moreover, during hot seasons, occupants tend to switch between different modes of cooling, including air conditioners, electric fans and natural ventilation. The penetration rates of HVAC systems and their properties should be further examined, since they share the greatest proportions of electricity consumption in buildings as shown in Section 3.3. Identifying the occupancy patterns and population densities should be a first basic step to enhance the reliability of UBEMs. Moreover, extensive periodic audits are required to investigate occupant-related parameters, such as economic status, internal thermal comfort, appliances usage, daily and seasonal occupancy schedules and activities. Efforts should be carried out to create an appropriate database for Beirut that incorporates construction materials and their thermal properties.

The calibration was employed at the temporal scale of the real data and yielded satisfactory accuracies for the overall building energy demand. However, the authors cannot claim accuracy at hourly level with no metered data for validation and/or calibration. Efforts should be placed on recording hourly electricity consumption by EDL or through verified occupant behavior modeling

[54–56]. The calibrated electricity consumption showed similar spatial clustering as the metered data from EDL in both districts, as was found in Section 3.4. This proves the relevance of our calibration process. The energy model can be employed to relate the observations of electricity consumption at one location to those at other locations. The main interest in revealing spatial clustering is its importance in generating zones for smart grid distribution.

The bottom-up approach presented in this paper overcomes the limitations in [9,10,24,25,45,57,58]. Furthermore, its calibration concentrates on integrating the significant weight of occupant behaviors into the urban model. Its capability to handle the complexity of an urban area can be used to enhance the findings of the previous studies within the Mediterranean region. The adopted methodology can be replicated in any other districts or cities in Lebanon and the region. Its multi-scalability is recognized spatially when ranging from building to the city level, and temporally when ranging from hourly to yearly resolution. This allows for a spatiotemporal energy patterns analysis to allocate hot spots and peak times of energy demands. In this regard, 2D and 3D urban energy maps are very informative (Fig. A.6) and enable critical analysis.

Policy implications. For Lebanon, a country with chronic energy sector failings reflected by the widening deficit of power delivery and quality, this work offers three distinct policy connections that could help with resolving real and existing challenges:

- First, due to the unrefined and bi-monthly billing of electricity as well as the distorted electricity market in the country, EDL, the national power utility does not have an accurate account for consumption loads and their variations spatially and temporally. The developed model provides a powerful tool to understand better load variations, and thus plan future strategies and investments accordingly. For example, one of the main techno-policy challenges lies in estimating Lebanon's real demand for electricity and its growth rate in the future due to the complex interactions between EDL and operators of private diesel generators, some of which are metered and some remain largely unregulated; and the existence of suppressed demand that is due to EDLs failure to provide power around the clock. With the help of smart metering that EDL is starting to imple-

ment, this model can be used to scientifically calibrate, verify and estimate growing demand on a national and sub-national scales.

- Second, until EDL can provide electricity without any interruptions, power outages will continue to occur. Predicting electricity consumption patterns in Beirut's urban environment, as shown in this work, could ultimately help inform smart rationing of the electricity that provides an efficient mapping between loads and available supply capacity.
- Third, the developed model could be utilized to test various policy instruments and ideas that promote energy savings such as the implementation of building efficiency codes, installation of water heaters, etc. One particularly useful application is to test the potential of rooftop solar PV systems to meet demand, or part of it, during certain daily and yearly peak times. Separately, the authors are working on a PV supply study that can ultimately be coupled with this work to provide a real assessment of the potential of rooftop solar PV systems to provide cost savings as well as increase power coverage in Beirut.

Disclosure of conflicts of interest

The authors have no competing interests to declare.

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Appendix A

A.1. Distribution of buildings' functions and periods of construction in the study areas

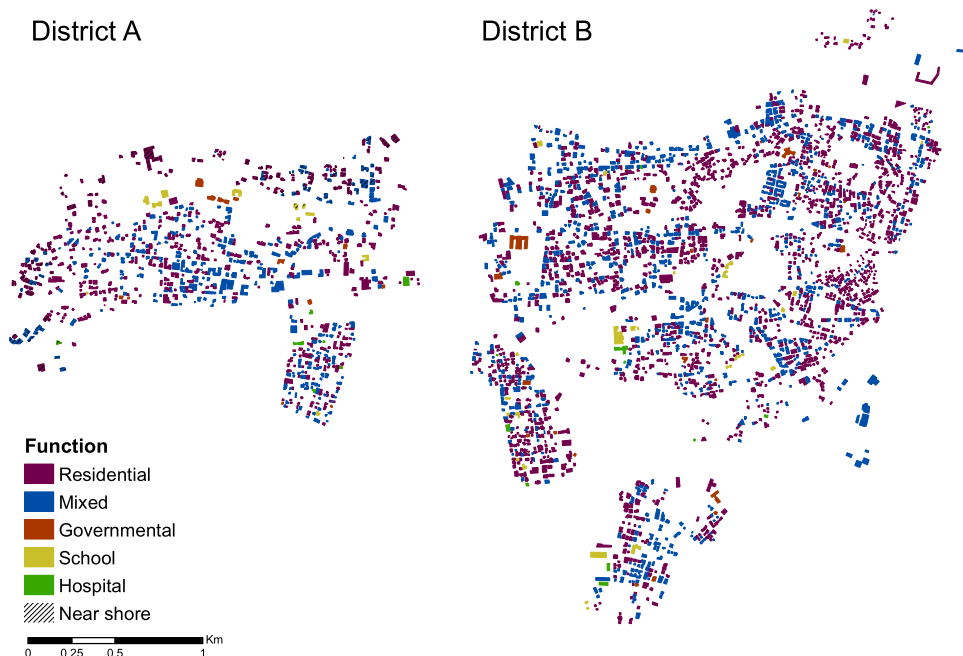


Fig. A.1. Buildings functions distribution in districts A and B.

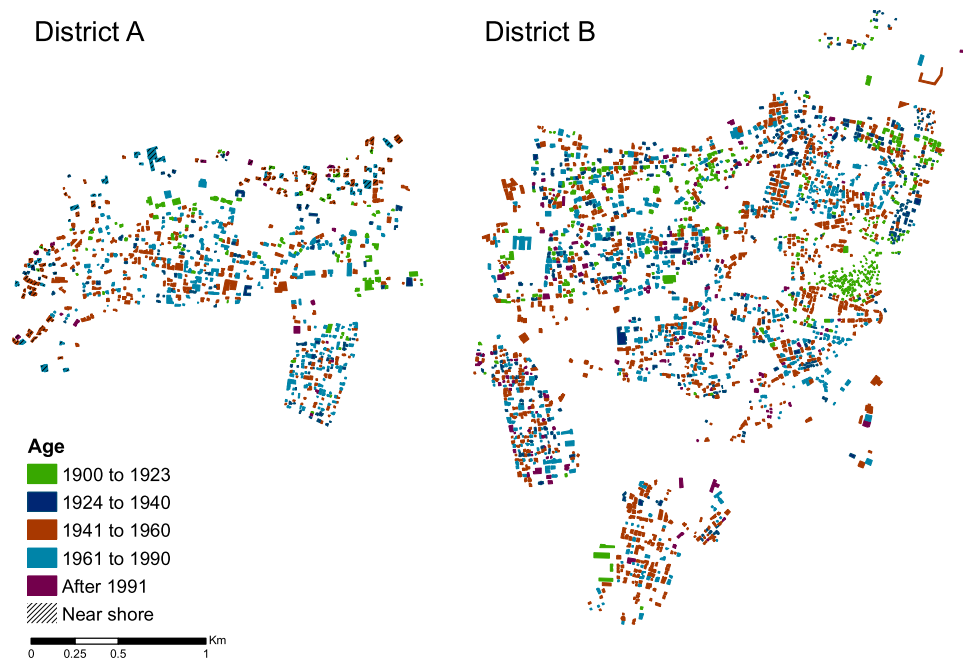


Fig. A.2. Buildings years of construction distribution in districts A and B.

A.2. Comparison of the total EDL electricity consumption with the simulated electricity demand before and after calibration

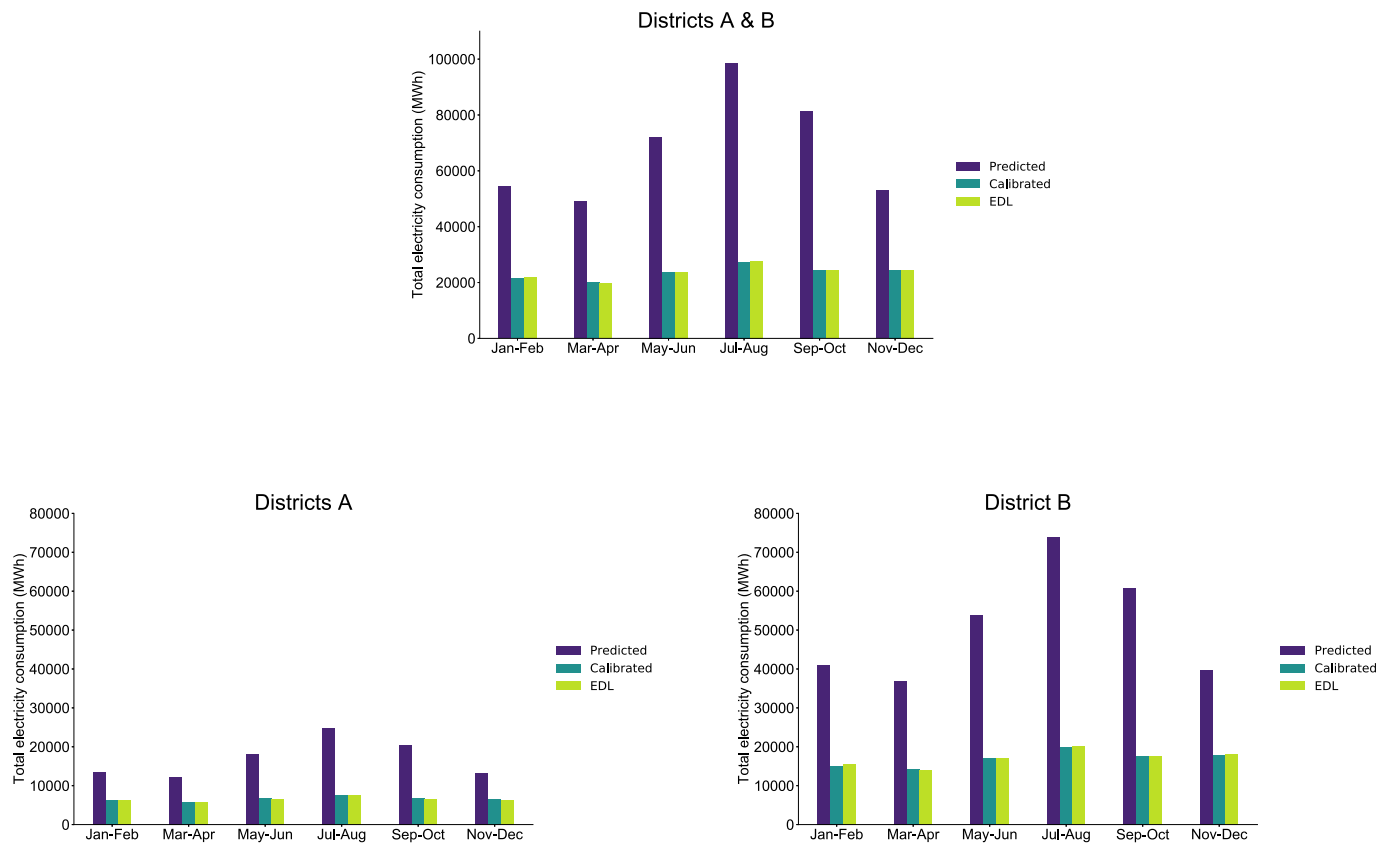


Fig. A.3. Total electricity demand in districts A and B.

A.3. Energy loads by end-uses by building type

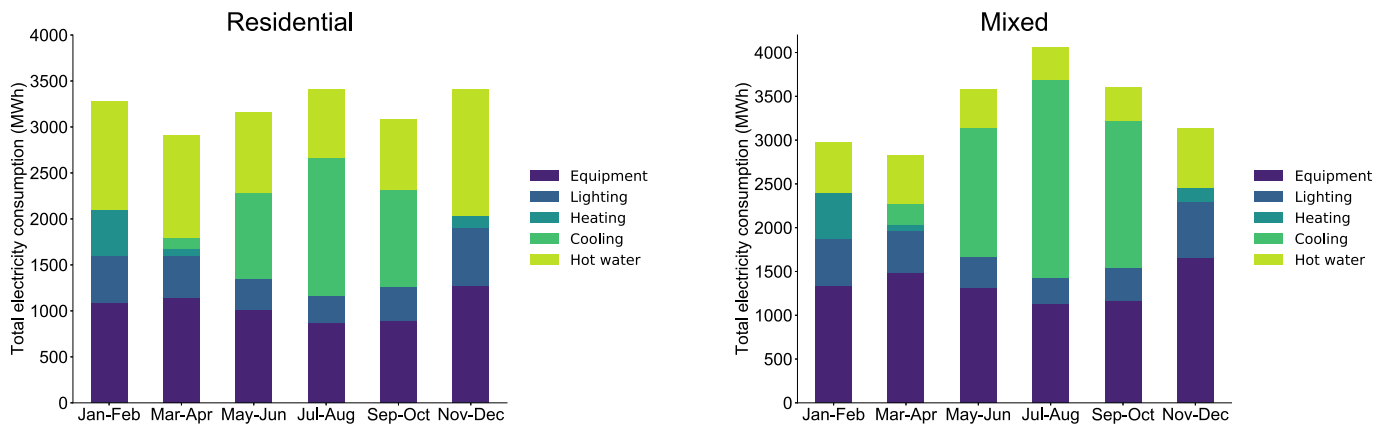


Fig. A.4. Bimonthly calibrated electricity consumption by buildings types in District A.

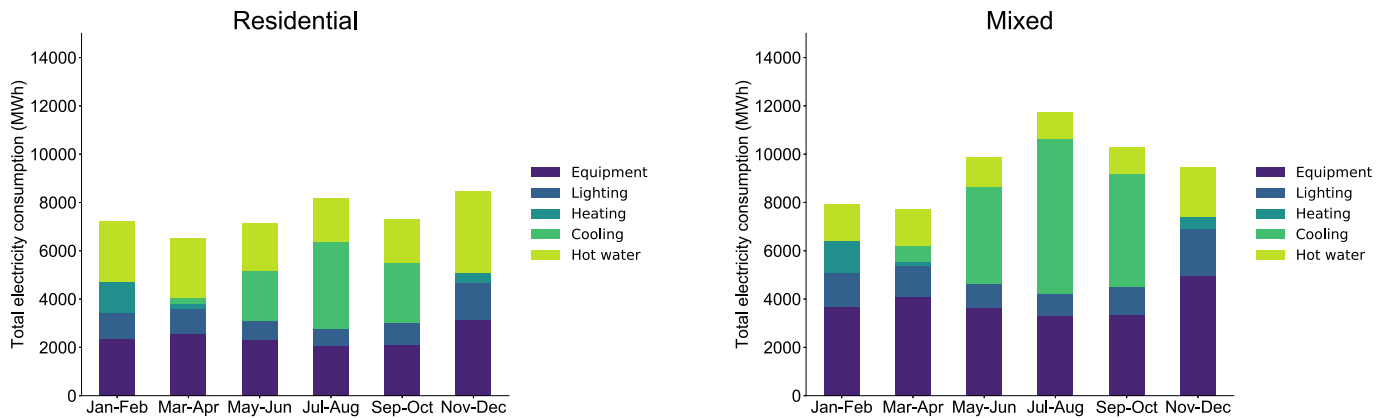


Fig. A.5. Bimonthly calibrated electricity consumption by buildings types in District B.

Table A.1

Percentage of energy consumption by end-uses for residential and mixed buildings in both districts.

District A		Jan-Feb	Mar-Apr	May-Jun	Jul-Aug	Sep-Oct	Nov-Dec	Yearly
Residential	Equipment	33	39	32	26	29	37	33
	Lighting	16	16	11	9	12	18	13
	Heating	15	3	0	0	0	4	4
	Cooling	0	4	29	44	34	0	19
	Hot water	36	38	28	22	25	40	31
Mixed	Equipment	45	53	37	28	32	53	40
	Lighting	18	17	10	8	11	21	13
	Heating	15	2	0	0	0	5	4
	Cooling	0	9	41	56	47	0	28
	Hot water	20	20	12	9	11	22	15
District B		Jan-Feb	Mar-Apr	May-Jun	Jul-Aug	Sep-Oct	Nov-Dec	Yearly
Residential	Equipment	32	39	32	25	29	37	33
	Lighting	15	16	11	9	12	18	13
	Heating	18	3	0	0	0	5	4
	Cooling	0	4	29	44	34	0	19
	Hot water	35	38	28	22	25	40	31
Mixed	Equipment	47	53	37	28	33	53	41
	Lighting	18	17	10	8	11	21	13
	Heating	17	2	0	0	0	5	3
	Cooling	0	8	41	55	46	0	28
	Hot water	19	20	12	9	11	22	15

A.4. Spatiotemporal distribution of the electricity consumption

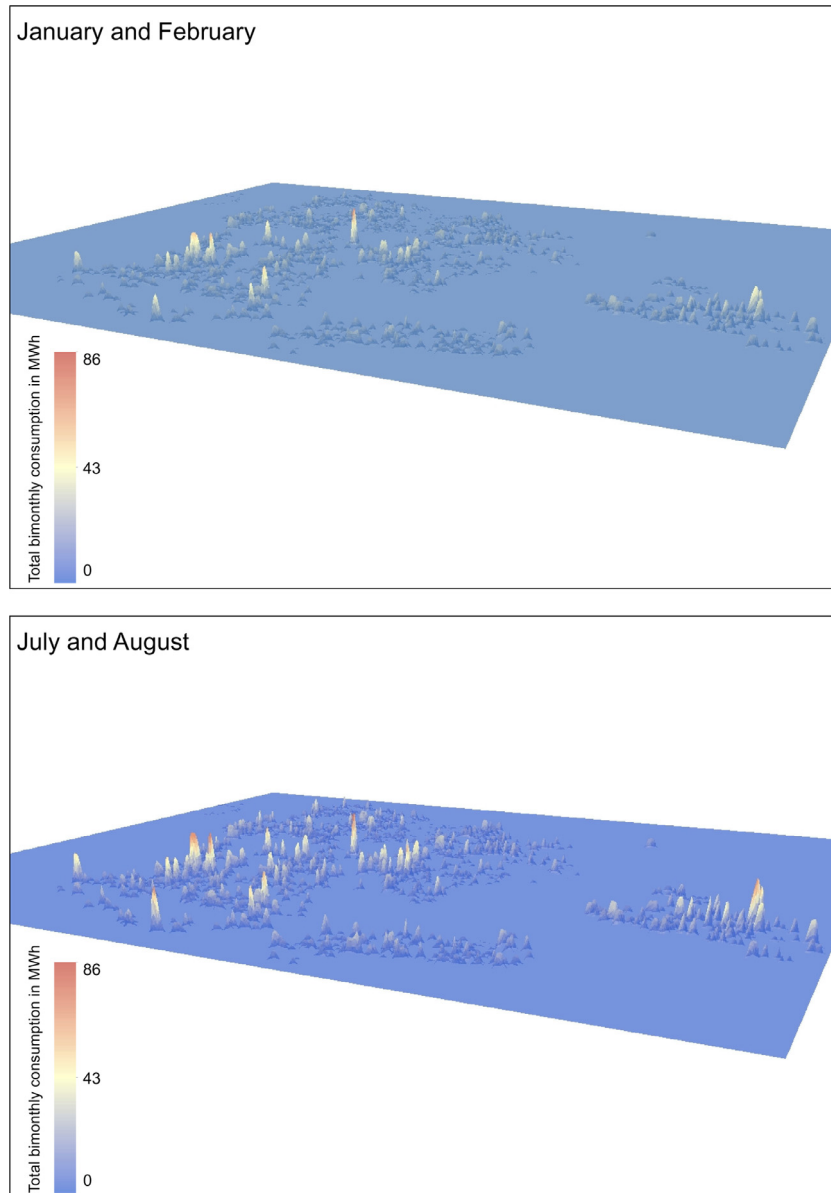


Fig. A.6. Spatiotemporal distribution of the bimonthly predicted electricity consumption of the buildings in district B.

A.5. Spatial correlation of buildings archetypes in terms of energy consumption

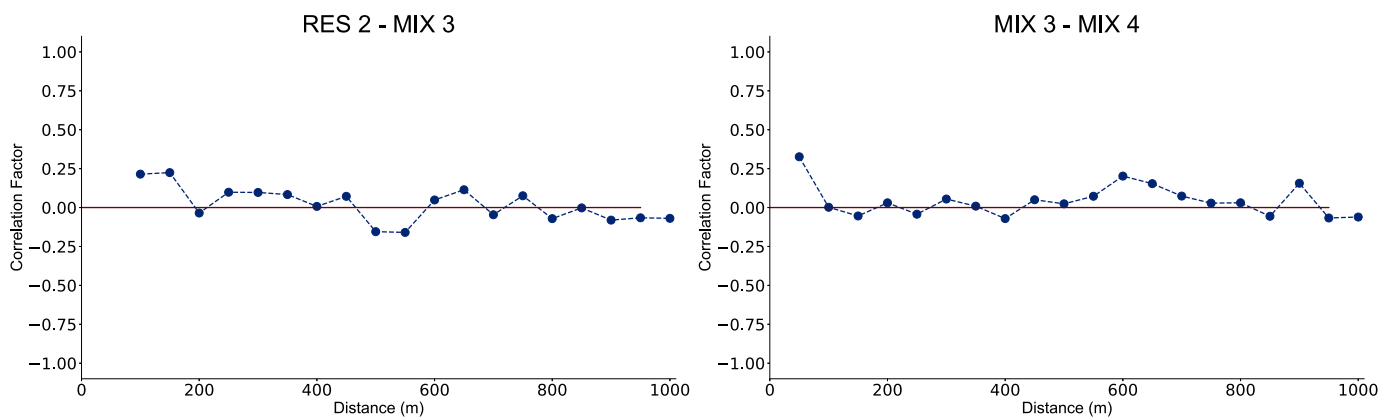


Fig. A.7. Spatial correlation between energy use of different buildings types in district A.

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